

ASSIGNMENT 1 OF PROBABILITY AND STATISTICS

1. There are 10 contestants for the post of chairman, secretary, treasurer and security. These positions can be filled by any of the 10 people. Find the possible number of ways in which the 4 positions may be filled.
2. The probability that a regularly scheduled flight departs on time is $P(B) = 0.83$; the probability that it arrives on time is $P(A) = 0.82$; and the probability that it departs and arrives on time is $P(A \cap B) = 0.78$. Find the probability that a plane
 - (a) arrives on time given that it departed on time,
 - (b) departed on time given that it has arrived on time.
3. Events A and B are such that: $P(A) = \frac{1}{2}$, $P(A/B) = \frac{1}{3}$ and $P(B/A) = \frac{3}{5}$.
 - i. Are A and B independent events?
 - ii. Are they mutually exclusive events?
 - iii. Find $P(A \cap B)$
 - iv. Find $P(B)$
4. If two events, A and B , are such that $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \cap B) = 0.2$, find the following:
 - (a) $P(A|B)$
 - (b) $P(B|A)$
 - (c) $P(A|A \cap B)$
 - (d) $P(A|A \cup B)$
 - (e) $P(A \cap B|A \cup B)$
5. In a certain assembly plant, three machines H_1, H_2 , and H_3 , make 30%, 45%, and 25%, respectively of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now suppose that a finished product is randomly selected.
 - (a) What is the probability that is defective?
 - (b) If a product were chosen randomly and found to be defective, what is the probability that it was made by machine H_3 ?
6. Find the expectation value, Variance, and Standard Deviation for the following probability distribution.

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

7. Find the value of k , expectation value, Variance, and Standard Deviation for the following probability distribution.

X	1	2	3	4	5	6
$P(X = x)$	0.1	0.2	0.24	k	0.04	0.05

8. A random variable X has a probability density function

$$f(x) = \begin{cases} \frac{k}{4}(x^2 + 1) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k , expectation, variance, standard deviation and mode.

9. A random variable X has a probability density function

$$f(x) = \begin{cases} kx^3 & \text{if } 2 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of k , expectation, variance and standard deviation.